

Here are conceptual answers to each question without using mathematical formulas.

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### **1. What problem do traditional Recurrent Neural Networks face when learning long sequences, and how do Long Short-Term Memories solve it?**

Traditional Recurrent Neural Networks (RNNs) struggle to remember information from the distant past when processing long sequences. As information passes through many time steps, important details gradually become weaker or disappear. This makes it difficult for an RNN to learn relationships between events that are far apart in a sequence. For example, in a long sentence, an RNN may forget the subject before reaching the verb.

Long Short-Term Memory (LSTM) networks solve this problem by introducing a memory mechanism that can preserve important information for a long time. Instead of treating all information equally, an LSTM learns what information should be kept, what should be discarded, and what new information should be added. This selective memory allows it to capture long-term dependencies much more effectively than a traditional RNN.

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### **2. Explain the difference between cell state and hidden state in an LSTM. Why are both needed?**

The **cell state** is the LSTM's long-term memory. It carries important information through the sequence with only small changes, allowing the network to remember relevant information over many time steps.

The **hidden state** is the short-term memory or the current output of the LSTM. It contains information that is most useful for the current time step and is passed to the next layer or used to make predictions.

Both are needed because they serve different purposes. The cell state preserves long-term information, while the hidden state focuses on the information that is currently relevant. Separating long-term memory from short-term output allows the LSTM to remember important past information without overwhelming the current computation.

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### **3. What is the role of the Forget Gate in an LSTM? What problems could happen if it did not exist?**

The **Forget Gate** decides which information in the cell state is no longer useful and should be removed. It helps the LSTM discard outdated or irrelevant information while keeping valuable memories.

If the Forget Gate did not exist, the LSTM would continue storing all past information, even if it was no longer relevant. Over time, the memory would become cluttered with unnecessary details, making it harder for the network to focus on important information. This could reduce prediction accuracy, increase confusion, and make learning long sequences much less effective.

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#### **4. How does the Input Gate help an LSTM decide what new information should be stored in memory?**

The **Input Gate** controls which new information should be added to the cell state. It examines the current input together with the previous hidden state and determines whether the new information is important enough to remember.

Only the information considered useful is stored in the cell state, while unimportant information is ignored. This selective process prevents the memory from being filled with irrelevant details and helps the LSTM maintain meaningful long-term information.

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#### **5. What is the purpose of the Output Gate and how does it help create the hidden state?**

The **Output Gate** determines which parts of the cell state's information should be revealed as the hidden state at the current time step.

Although the cell state may contain a large amount of stored information, not all of it is needed immediately. The Output Gate selects the most relevant information for the current context and uses it to produce the hidden state. This hidden state is then passed to the next time step and can also be used to make predictions.

By controlling what information is exposed, the Output Gate ensures that the hidden state contains only the most useful information for the current task while the remaining information stays safely stored in the cell state for future use.

**\*\*\*Shortly Describe the working flow---**

#### **LSTM Working Flow (Short Description)**

- 1. Receive Input:** The LSTM receives the current input along with information from the previous time step.
- 2. Forget Gate:** It decides what old information in the memory is no longer useful and should be removed.
- 3. Input Gate:** It determines what new information is important and stores it in the memory.
- 4. Update Cell State:** The memory (cell state) is updated by removing unnecessary information and adding the selected new information.
- 5. Output Gate:** It selects the relevant information from the updated memory to produce the hidden state.
- 6. Pass to Next Step:** The updated cell state (long-term memory) and hidden state (current output) are passed to the next time step, where the process repeats.

**In summary:**

**Input → Forget Gate → Input Gate → Update Cell State → Output Gate → Hidden State → Next Time Step**